

KUGGOREN, SwedenWMO: **023550**Lat: **61.703N**Lon: **17.522E**Elev: **10**StdP: **101.20**Time Zone: **1.00 (EUC)**Period: **96-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.7	-11.1	-19.9	0.6	-7.9	-16.5	0.9	-9.0	17.4	-1.3	15.8	-1.1	4.4	320	0.706

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.6	22.2	17.8	20.9	17.0	19.6	16.2	18.9	21.2	17.9	19.9	17.0	18.9	4.1	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.9	12.9	20.3	17.0	12.2	19.2	16.0	11.4	18.2	53.6	21.2	50.5	20.0	47.6	19.0	22.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	14.1	12.2	10.5	-16.1	25.4	3.3	1.9	-18.5	26.8	-20.4	27.9	-22.3	29.0	-24.7	30.3	
(5)				-16.9	19.4	3.0	1.3	-19.0	20.4	-20.8	21.1	-22.4	21.9	-24.6	22.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	6.1	-2.0	-1.9	0.2	3.5	7.8	12.5	16.3	15.8	11.8	6.5	2.3	-0.4	(6)
(7)		DBStd	7.28	4.30	4.56	4.00	3.03	2.88	2.55	2.34	2.29	2.64	2.86	3.26	4.10	(7)
(8)		HDD10.0	1971	372	334	304	197	80	6	0	0	12	112	231	323	(8)
(9)		HDD18.3	4488	630	567	562	446	326	177	73	84	194	367	481	582	(9)
(10)		CDD10.0	539	0	0	0	1	13	80	195	180	67	4	0	0	(10)
(11)		CDD18.3	16	0	0	0	0	0	1	9	6	0	0	0	0	(11)
(12)		CDH23.3	19	0	0	0	0	0	1	13	5	0	0	0	0	(12)
(13)		CDH26.7	2	0	0	0	0	0	0	2	1	0	0	0	0	(13)
(14)	Wind	WSAvg	4.9	5.6	5.5	5.0	4.5	4.4	4.5	4.2	4.3	4.5	5.2	5.4	5.4	(14)
(15)	Precipitation	PrecAvg	458	26	23	17	27	38	40	51	68	38	50	44	36	(15)
(16)		PrecMax	693	68	35	48	70	67	71	122	139	75	121	110	81	(16)
(17)		PrecMin	311	2	10	0	4	5	0	14	11	13	0	12	5	(17)
(18)		PrecStd	90	17	8	11	19	19	22	30	32	17	35	27	17	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.2	8.3	11.5	15.0	18.3	21.7	24.8	23.6	20.9	14.7	11.2	9.0	(19)
(20)		MCWB	3.8	4.5	6.1	9.0	10.3	14.7	18.3	17.8	15.2	11.5	8.3	6.4		(20)
(21)		2%	DB	5.3	6.8	9.1	11.6	15.3	19.2	22.8	21.8	18.1	12.6	8.9	7.0	(21)
(22)		MCWB	2.8	3.6	4.6	6.6	10.1	13.9	18.3	17.7	14.0	10.5	6.9	5.0		(22)
(23)		5%	DB	4.2	5.3	7.4	9.5	13.7	17.6	21.4	20.5	16.8	11.6	7.4	5.7	(23)
(24)		MCWB	2.2	2.8	3.8	5.5	9.7	13.2	17.8	17.1	13.6	9.8	6.1	4.0		(24)
(25)		10%	DB	3.1	3.5	5.8	7.7	12.1	16.3	20.0	19.4	15.6	10.4	6.4	4.7	(25)
(26)		MCWB	1.6	1.5	2.7	4.6	8.6	12.6	16.8	16.4	13.0	8.7	5.5	3.2		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.7	5.2	6.8	9.1	12.4	16.2	20.6	20.0	16.3	12.4	9.0	7.1	(27)
(28)		MDCB	6.3	7.7	10.8	14.1	15.7	19.6	22.7	22.2	18.7	13.4	10.8	8.4		(28)
(29)		2%	WB	3.5	4.0	5.2	7.2	11.0	14.9	19.4	18.7	15.2	11.2	7.4	5.5	(29)
(30)		MDCB	4.7	6.3	8.4	10.6	14.3	17.9	21.7	20.8	17.2	12.3	8.5	6.6		(30)
(31)		5%	WB	2.7	3.0	4.0	5.9	9.9	14.0	18.3	17.9	14.3	10.1	6.5	4.4	(31)
(32)		MDCB	3.7	5.0	7.1	8.7	13.0	16.7	20.6	19.7	15.9	11.4	7.2	5.5		(32)
(33)		10%	WB	1.9	1.9	2.8	4.9	9.0	13.2	17.2	17.1	13.7	8.8	5.7	3.4	(33)
(34)		MDCB	2.8	3.3	5.4	7.3	11.6	15.6	19.4	18.9	15.1	10.0	6.3	4.4		(34)
(35)	Mean Daily Temperature Range	MDBR	4.1	4.3	4.7	4.8	4.8	4.8	4.6	4.6	4.7	3.9	3.3	3.9		(35)
(36)		5% DB	MCDBR	4.6	5.5	7.2	8.6	8.0	7.4	6.6	6.0	6.5	5.2	4.3	4.6	(36)
(37)		MCWBR	3.8	3.9	4.5	4.9	4.3	3.9	3.6	3.5	4.0	3.9	3.8	3.9		(37)
(38)		5% WB	MCDBR	4.0	5.3	6.9	8.0	7.2	6.4	5.3	4.7	5.2	4.6	4.0	4.2	(38)
(39)		MCWBR	3.5	4.1	4.6	4.9	4.3	3.9	3.5	3.3	3.7	3.9	3.7	3.8		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.253	0.304	0.324	0.359	0.360	0.354	0.374	0.367	0.334	0.320	0.271	0.343		(40)
(41)		taud	2.169	2.283	2.373	2.372	2.437	2.494	2.458	2.471	2.544	2.437	2.208	2.580		(41)
(42)		Ebn at Noon	459	666	794	832	862	874	843	818	787	650	423	297		(42)
(43)		Edh at Noon	35	62	81	98	100	96	97	89	69	54	33	22		(43)
(44)	All-Sky Solar Radiation	RadAvg	0.23	0.86	2.37	3.85	4.92	5.58	5.21	3.96	2.51	1.13	0.33	0.13		(44)
(45)		RadStd	0.03	0.16	0.27	0.33	0.52	0.37	0.54	0.35	0.31	0.19	0.07	0.01		(45)